

Objective-Driven Sequential Bayesian Experimental Design for Power-Grid Dynamic Learning and Control

[TODO: Authors and affiliations]

Abstract—This paper studies sequential Bayesian experimental design for uncertain power-grid dynamics. At each stage, a policy selects a probing bus and duration, observes a noisy frequency response, and updates uncertainty over machine-specific parameters. We distinguish parameter-oriented expected information gain (EIG) from objective-driven mean objective cost of uncertainty (MOCU) for a downstream safety-aware control decision. Random, fixed, myopic, deep adaptive design (DAD), and reinforcement-learning sequential Bayesian optimal experimental design (RL-sBOED), and Step-DAD are evaluated under common simulation banks and held-out protocols. A susceptible–infected–recovered (SIR) model provides an implementation benchmark; IEEE 9-bus, IEEE 14-bus, and IEEE 30-bus systems assess the power-grid application and scalability. [TODO: Add final quantitative findings.]

Index Terms—Bayesian experimental design, power-system dynamics, adaptive design, mean objective cost of uncertainty, information gain.

I. INTRODUCTION

UNCERTAINTY in power-system inertia and fast frequency response complicates frequency-security assessment and control selection. Controlled probing can reduce uncertainty, but an experiment must determine where to probe, how long to probe, and whether later probes should depend on earlier observations. [TODO: Add motivation and citations.]

EIG values parameter learning, but parameter information does not necessarily improve a final control decision. We therefore also study MOCU, which values an experiment by its reduction of expected decision penalty.

The contributions are:

- sequential bus–duration BOED for uncertain power-grid dynamics;
- a safety-aware MOCU objective linking probing to terminal control;
- a matched comparison of random, fixed, myopic, DAD, RL-sBOED, and Step-DAD; and
- SIR, IEEE9, IEEE14, and IEEE30 studies of objective choice, adaptation, horizon, and scalability.

II. RELATED WORK

Summarize BOED and EIG, adaptive and non-myopic design, DAD, RL-sBOED, Step-DAD, MOCU, and power-grid dynamic identification. End with the gap: objective-driven sequential design over power-grid probing buses and durations. [TODO: Add verified references.]

[TODO: Funding and corresponding-author information.]

III. PROBLEM FORMULATION

A. Power-Grid Model

Let $\mathcal{N} = \{1, \dots, N\}$ denote all physical buses, and let $\mathcal{M} \subset \mathcal{N}$ denote the buses containing dynamic machines. The frequency response of those machines is modeled by

$$\mathbf{M}\dot{\boldsymbol{\omega}}(s) = -\Delta\mathbf{P}_e(s) - \left(\frac{\mathbf{K}}{2\pi} + \mathbf{D}\right)\boldsymbol{\omega}(s) + \mathbf{u}(s), \quad (1)$$

following the inertia–frequency–response model in [1]. The vectors $\boldsymbol{\omega}$ and $\Delta\mathbf{P}_e$ contain angular- frequency deviations and network electrical-power changes at the dynamic machines. The diagonal matrices $\mathbf{M}$, $\mathbf{K}$, and $\mathbf{D}$ contain inertia, fast frequency-response, and damping coefficients, respectively, and $\mathbf{u}$ is an external active-power injection.

Only the machine-specific coefficients are uncertain:

$$\boldsymbol{\theta} = ((M_i, K_i) : i \in \mathcal{M}) \in \Theta, \quad \boldsymbol{\theta} \sim p_0(\boldsymbol{\theta}). \quad (2)$$

Thus, every physical bus is an admissible probe location, whereas M_i and K_i are defined only for $i \in \mathcal{M}$. The IEEE9 and IEEE14 models have, respectively, $(N, |\mathcal{M}|) = (9, 3)$ and $(14, 5)$; their static network data follow the standard IEEE test cases distributed with MATPOWER [2].

Let $\mathbf{a}(b)$ map a scalar injection at physical bus $b \in \mathcal{N}$ to the dynamic-machine equations. At a machine bus, the injection enters that machine directly. At a non-machine bus, Kron reduction distributes the injection among the dynamic machines. A non-machine bus therefore remains a valid probe location without being assigned artificial dynamic states, M_i , or K_i .

B. Sequential Probe Design

At stage t , the design is a bus–duration pair

$$\xi_t = (b_t, \tau_t) \in \Xi = \mathcal{N} \times \mathcal{T}, \quad (3)$$

where $\mathcal{T}$ is the six-duration catalog and the amplitude A is fixed. The probing signal is the Hann pulse used for controlled frequency excitation in [1]:

$$u_p(s; \tau_t) = \begin{cases} \frac{A}{2} \left[1 - \cos\left(\frac{2\pi s}{\tau_t}\right) \right], & 0 \leq s \leq \tau_t, \\ 0, & s > \tau_t. \end{cases} \quad (4)$$

During a probe simulation,

$$\mathbf{u}(s) = \mathbf{a}(b_t)u_p(s; \tau_t), \quad u_{\text{ctrl}} = 0, \quad (5)$$

and the system is reset to its configured equilibrium before the next probe. IEEE9 and IEEE14 contain $9 \times 6 = 54$ and $14 \times 6 = 84$ actions, respectively. Exact repetition of a bus–duration pair is disallowed within a rollout.

After t probes, the realized history is

$$h_t = ((\xi_1, \mathbf{y}_1), \dots, (\xi_t, \mathbf{y}_t)). \quad (6)$$

An adaptive policy selects

$$\xi_t \sim \pi_t(\cdot \mid h_{t-1}), \quad t = 1, \dots, T, \quad (7)$$

subject to the no-repetition constraint. This is the standard adaptive formulation of sequential BOED [3]. The horizon T is the probe budget. A fixed design selects its entire sequence before any data are observed; an adaptive design may change later actions in response to the realized observations.

C. Observation Model and Bayesian Update

Let $o \in \mathcal{M}$ be the fixed PMU bus and $\Delta f_o(s) = \omega_o(s)/(2\pi)$ its frequency deviation. For sampling set $\mathcal{S} = \{s_1, \dots, s_{N_{\text{obs}}}\}$, the simulator output is

$$\mathbf{g}(\boldsymbol{\theta}, \xi_t) = [\Delta f_o(s; \boldsymbol{\theta}, \xi_t)]_{s \in \mathcal{S}}. \quad (8)$$

The measured response is

$$\mathbf{y}_t = \mathbf{g}(\boldsymbol{\theta}^*, \xi_t) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \sigma_y^2 \mathbf{I}). \quad (9)$$

The main power-grid experiments use $N_{\text{obs}} = 5$. Because every probe starts from equilibrium, the responses are conditionally independent given $\boldsymbol{\theta}$ and the selected designs:

$$p(\boldsymbol{\theta} \mid h_t) \propto p_0(\boldsymbol{\theta}) \prod_{k=1}^t p(\mathbf{y}_k \mid \boldsymbol{\theta}, \xi_k). \quad (10)$$

The implementation represents this posterior with N_p particles $\{\boldsymbol{\theta}^{(n)}, w_n^{(t)}\}_{n=1}^{N_p}$. Their weights are updated by

$$w_n^{(t)} = \frac{w_n^{(t-1)} p(\mathbf{y}_t \mid \boldsymbol{\theta}^{(n)}, \xi_t)}{\sum_{j=1}^{N_p} w_j^{(t-1)} p(\mathbf{y}_t \mid \boldsymbol{\theta}^{(j)}, \xi_t)}. \quad (11)$$

D. Parameter-Oriented Design Objective

Let H_T denote the random terminal history induced by policy π . For the parameter-learning experiments, the policy maximizes

$$J_{\text{EIG}}(\pi) = \mathbb{E}_{H_T \mid \pi} [D_{\text{KL}}(p(\boldsymbol{\theta} \mid H_T) \parallel p_0(\boldsymbol{\theta}))]. \quad (12)$$

This is the expected information gain used in sequential BOED [3], [4]. It rewards posterior learning about $\boldsymbol{\theta}$ but does not directly encode the quality of the downstream control decision.

E. Terminal Control and Safety

Diagnostic probing and terminal-control evaluation are separate simulations. In the latter, the probe is zero and

$$\mathbf{u}(s) = \mathbf{a}(b_c)u_c(s) + \mathbf{a}(b_u)u h_{\text{ctrl}}(s), \quad u_p = 0, \quad (13)$$

where u_c is the fixed contingency, $u \in \mathcal{U}$ is a candidate supplementary control magnitude, and b_c and b_u are their configured locations.

For parameter $\boldsymbol{\theta}$ and control u , let $r(\boldsymbol{\theta}, u)$ be the maximum absolute rate of change of frequency (ROCOF), and let $f_{\min}(\boldsymbol{\theta}, u)$ be the minimum frequency deviation. A control is safe when

$$\text{Safe}(\boldsymbol{\theta}, u) \iff r(\boldsymbol{\theta}, u) \leq r_{\max} \text{ and } f_{\min}(\boldsymbol{\theta}, u) \geq \underline{f}. \quad (14)$$

For the ordered admissible grid $\mathcal{U}$, define the required control

$$U(\boldsymbol{\theta}) = \min\{u \in \mathcal{U} : \text{Safe}(\boldsymbol{\theta}, u)\}. \quad (15)$$

The retained parameter support is required to contain a safe candidate. The offline bank stores $U_n = U(\boldsymbol{\theta}^{(n)})$ for each particle.

Given terminal weights $\mathbf{w}^{(T)}$, the deployed control is

$$u_T = \left\lceil Q_{1-\alpha} \left(U \mid \mathbf{w}^{(T)} \right) + m \right\rceil_{\mathcal{U}}. \quad (16)$$

Here Q_q is the weighted empirical q -quantile, m is a safety margin, and $\lceil \cdot \rceil_{\mathcal{U}}$ rounds upward to the control grid. The main MOCU experiments use $\alpha = 0.05$ and $m = 0$, corresponding to a 95% posterior-safety target.

F. Objective-Driven Design Objective

MOCU measures the expected performance loss caused by uncertainty relative to model-specific operation [5]. For a system requiring control U , define the safety-aware operational cost

$$L(u, U) = u + \lambda(U - u)_+ + \kappa \mathbf{1}\{U > u\}, \quad (17)$$

where $(x)_+ = \max(x, 0)$, λ penalizes control shortfall, and κ optionally adds a discrete safety-violation penalty. Since the model-specific decision is $u = U$, its cost is $L(U, U) = U$. The objective cost of uncertainty for particle n is

$$\text{OCU}_n(h_T) = L(u_T, U_n) - U_n, \quad (18)$$

and the posterior MOCU is

$$\text{MOCU}(h_T) = \sum_{n=1}^{N_p} w_n^{(T)} [L(u_T, U_n) - U_n]. \quad (19)$$

This is a frequency-safety specialization of MOCU-based uncertainty quantification and experimental design [5], [6].

The implementation sets $\lambda = 1/\alpha = 20$ and $\kappa = 0$. Under this loss, the Bayes-optimal continuous decision is the $(1 - \alpha)$ quantile of U ; rounding it upward gives (16). Thus, the deployed control and the reported MOCU solve the same terminal decision problem.

The objective-driven policy is

$$\pi_{\text{MOCU}}^* = \arg \min_{\pi} \mathbb{E}_{H_T \mid \pi} [\text{MOCU}(H_T)], \quad |H_T| = T. \quad (20)$$

Lower terminal MOCU is better. Unlike EIG, this objective values parameter learning only when it improves the downstream control decision.

TABLE I
COMPARED METHODS

Method	Adaptive	Non-myopic	Learned
Random	No	No	No
Fixed	No	Open-loop	No
Myopic	Yes	No	No
DAD	Yes	Yes	Yes
RL-sBOED	Yes	Yes	Yes
Step-DAD	Yes	Yes	Semi-amortized

IV. COMPARED DESIGN METHODS

Random ignores observations. Fixed selects an open-loop sequence. Myopic optimizes immediate expected utility after each observation. DAD learns an amortized history-conditioned policy offline. RL-sBOED learns a sequential policy with stepwise rewards. [TODO: Give reproducible implementation details.] Step-DAD starts from an amortized DAD policy and performs limited history-specific online refinement. Report its offline DAD training cost, online refinement cost, and action-selection cost separately. [TODO: Describe the implementation and deviations from the original paper.]

V. EXPERIMENTAL SETUP

A. SIR EIG Benchmark

SIR verifies the EIG implementation on a simpler nonlinear model; it is not the application contribution. [TODO: State its parameters, actions, and noise.]

B. IEEE9 Experiments

IEEE9 has nine physical probe buses, three dynamic machines, and six uncertain M_i/K_i coordinates. Six durations give 54 actions. Evaluate EIG and MOCU with $N_{\text{obs}} = 5$, $\sigma = 0.005$, $T = 3, \dots, 7$, and seeds 101, 202, and 303. [TODO: Insert final durations and priors.]

C. IEEE14 MOCU Extension

IEEE14 has 14 physical probe buses, five dynamic machines, and ten uncertain M_i/K_i coordinates. Six durations give 84 actions. The MOCU workflow, methods, noise, horizons, and seeds match IEEE9; only IEEE-system physics and machine parameters differ. [TODO: Insert the validated durations.]

D. IEEE30 MOCU Scalability Study

IEEE30 extends the same objective-driven MOCU protocol to a larger network. Use machine-specific uncertain parameters only for its dynamic machines, allow probing at all configured physical buses, and retain the same methods, noise, horizons, seeds, terminal objective, and held-out evaluation rules used for IEEE9 and IEEE14. [TODO: Insert the verified dynamic-machine mapping, prior ranges, action count, bank size, and audited six-duration combination.]

E. Duration Selection and Evaluation

Select six durations before policy training using a method-independent search over 281 values from 0.20 to 3.00 s. Screen adaptive value, non-myopic value, and observation-dependent branching; do not tune durations for a reported method. All methods share physical banks, held-out systems, and noise realizations. Report three-seed mean and sample standard deviation, plus paired held-out differences. [TODO: Add bank sizes and evaluation sample counts.]

VI. RESULTS

A. SIR Implementation Validation

[TODO: Report EIG and runtime for every method.]

B. IEEE9 EIG Results

[TODO: Report mean $\pm$ standard deviation for $T = 3, \dots, 7$.]

C. IEEE9 MOCU Results

[TODO: Report terminal MOCU, safety, paired differences, and controls.]

D. IEEE14 MOCU Results

[TODO: Report the same methods and metrics as IEEE9.]

E. IEEE30 MOCU Results

[TODO: Report the same methods and metrics as IEEE9 and IEEE14, emphasizing scalability in action count, latent dimension, training time, and online time.]

F. Computational Cost

[TODO: Separate bank generation, offline training, online selection, and Step-DAD history-specific refinement time.]

VII. DISCUSSION

Discuss when adaptation and non-myopic planning improve the downstream objective, why Fixed can be competitive, why EIG and MOCU can disagree, and how horizon and system size affect performance. Limitations include simulated Kron-reduced networks, prescribed priors, fixed probe amplitude, discrete actions, simplified terminal control, and three training seeds. Machine-specific K_i allocations introduced for the larger IEEE systems are modeling assumptions unless directly supported by a cited dynamic-data source.

VIII. CONCLUSION

This paper formulates power-grid probing as sequential BOED and connects learning to safety-aware terminal control through MOCU. [TODO: State only final conclusions supported by completed three-seed results.]

ACKNOWLEDGMENT

[TODO: Add funding and computing resources.]

TABLE II
TERMINAL MOCU ACROSS THREE SEEDS (MEAN $\pm$ STANDARD DEVIATION)

Method	$T = 3$	$T = 4$	$T = 5$	$T = 6$	$T = 7$
Random	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Fixed	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Myopic	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
DAD	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RL-sBOED	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Step-DAD	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

REFERENCES

- [1] J. Peng, J. Tan, P. Koralewicz, E. Mendiola, S. Dong, A. Hoke, and K. Horowitz, “Probing signal-based inertia and frequency response estimation for power systems with high levels of inverter-based resources,” in *2024 IEEE Power & Energy Society General Meeting (PESGM)*, 2024, pp. 1–5.
- [2] R. D. Zimmerman, C. E. Murillo-Sanchez, and R. J. Thomas, “MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education,” *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.
- [3] A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth, “Deep adaptive design: Amortizing sequential bayesian experimental design,” in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 3384–3395.
- [4] T. Blau, E. V. Bonilla, I. Chades, and A. Dezfouli, “Optimizing sequential experimental design with deep reinforcement learning,” in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 2107–2128.
- [5] B.-J. Yoon, X. Qian, and E. R. Dougherty, “Quantifying the objective cost of uncertainty in complex dynamical systems,” *IEEE Transactions on Signal Processing*, vol. 61, no. 9, pp. 2256–2266, 2013.
- [6] Q. Chen, X. Chen, H.-M. Woo, and B.-J. Yoon, “Neural message-passing for objective-based uncertainty quantification and optimal experimental design,” *Engineering Applications of Artificial Intelligence*, vol. 123, p. 106171, 2023.